

Building a system, learning to understand it

Xuanye(Sawyer) Xu

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01 My role and contribution

I was responsible for developing the simulation and hardware programs, physical debugging, and guidance on the grasping scene. I participated throughout the project, from defining the fixed-base bottle task to refining target positions and preparing submission materials.

During testing, I positioned the bottle, operated the terminal, observed the arm, and compared its behaviour with logs. I helped determine water quantity, pickup and placement points, safe height, and the home-pick-place-return sequence. AI assisted with code generation and revision; I contributed requirements, physical observations, verification, and decisions about proposed changes.

MY CONTRIBUTION

Define, observe, verify, and refine.

02 Improving the way I worked

Connecting to the Jetson through SSH and VNC allowed development from my Mac. SSH supported terminal operations, while VNC provided the graphical desktop. This reduced repeated monitor and mouse setup and made development less dependent on a fixed workstation. It showed me that efficiency also depends on removing repetitive preparation, not simply producing code faster.

REMOTE DEVELOPMENT

Less setup, greater flexibility.

03 Code and understanding

AI could generate files faster than I could understand them. Unfamiliar ROS 2 concepts, MoveIt motion planning, and communication settings sometimes made the code feel like a foreign language, leading to frustration and self-doubt.

The gripper-release problem illustrated why understanding matters. The visible symptom was a closed gripper, but logs revealed that an earlier movement had failed, preventing the release step. I learned to distinguish planning, command execution, and physical feedback. AI could suggest explanations and changes, but reading, remembering, testing, and connecting concepts remained my responsibility.

THE CENTRAL LESSON

Generating code does not replace understanding it.

04 From prototype to product

I wanted a complete product that ordinary users could start with a click. The result remained an experimental prototype. Time, energy, learning speed, and team coordination limited progress, but my own tendency to postpone difficult learning and documentation also mattered.

Sometimes I preferred seeing a function run to understanding it thoroughly. Next time, I would first understand a minimal working system, document decisions as they happen, and design setup, controls, and error recovery earlier. My main gain was recognising that effective tool use requires ownership of both the system and my learning.

NEXT TIME

Understand first. Document continuously. Design for users.